

The mean

The mean is the sum of all the values in a set of data divided by the number of values in the list. It is what most people understand by the word "average." To find the mean, a simple formula is used.

$$\text{Mean} = \frac{\text{Sum total of values}}{\text{Number of values}}$$

formula to find mean

First, take the list of data and put it in order. Count the number of values in the list. In this example, there are five values.

150, 160, 170, 180, 180

there are five numbers in list

add numbers together

sum total of values

Add all of the values in the list together to find the sum total of the values. In this example the sum total is 840.

$$150 + 160 + 170 + 180 + 180 = 840$$

Divide the sum total of the values, in this case 840, by the number of values, which is 5. The answer, 168, is the mean value of the list.

$$\frac{840}{5} = 168$$

sum total of values

number of values

168 is the mean

The median

The median is the middle value in a set of data. In a list of five values, it is the third value. In a list of seven values, it would be the fourth value.

median is middle value, in this case the orange figure



Firstly, put the data in ascending order (from lowest to highest)

170, 180, 180, 160, 150

The median is the middle value in a list with an odd number of values.

in this list of five values, third value is the median

150, 160, 170, 180, 180

LOOKING CLOSER

Median of an even number of values

In a list with an even number of values, the median is worked out using the two middle values. In a list of six values, these are the third and fourth values.

3rd value

4th value

middle values

150, 160, 170, 180, 180, 190

▷ **Calculating the median**
Add the two middle values and divide by two to find the median.

$$\frac{170 + 180}{2} = \frac{350}{2} = 175$$

median value

